

Question: 1

In Cisco Packet Tracer, create a simple two-switch network with three VLANs (e.g., VLAN 10 for 'Foodies', VLAN 20 for 'Gamers', VLAN 30 for 'Shoppers'), assign three PCs to each VLAN, and verify that PCs in the same VLAN can ping each other but not PCs in other VLANs?

Answer: 1. Topology Setup & Hardware Connections

Switches: Add two 2960 Switches (Switch1 and Switch2) to your workspace.

PCs: Add nine PCs in total (4 or 5 connected to Switch1, and the remaining connected to Switch2).

Inter-Switch Connection: Connect Switch1 and Switch2 together using a copper cross-over cable plugged into port FastEthernet 0/24 on both ends.

End-Device Connections: Connect the PCs to ports Fa0/1 through Fa0/9 on each switch.

2. Base IP Addressing Scheme (Same Subnet per VLAN)

Assign static IP addresses to the PCs in their desktop configuration windows matching these subnet groups:

VLAN 10 (Foodies): Use IPs 192.168.10.X (Subnet Mask: 255.255.255.0)

VLAN 20 (Gamers): Use IPs 192.168.20.X (Subnet Mask: 255.255.255.0)

VLAN 30 (Shoppers): Use IPs 192.168.30.X (Subnet Mask: 255.255.255.0)

3. Switch Configuration (Run on BOTH Switch1 & Switch2)

Open the CLI tab on each switch and paste this exact execution block to build the VLANs, assign ports, and activate the trunk link between the switches:

text

```
enable
configure terminal
```

```
vlan 10
  name Foodies
vlan 20
  name Gamers
vlan 30
  name Shoppers
exit
```

```
interface range FastEthernet 0/1 - 3
  switchport mode access
  switchport access vlan 10
exit
```

```
interface range FastEthernet 0/4 - 6
  switchport mode access
  switchport access vlan 20
exit
```

```
interface range FastEthernet 0/7 - 9
  switchport mode access
  switchport access vlan 30
exit
```

```
interface FastEthernet 0/24
  switchport mode trunk
exit
```

Use code with caution.

4. Verification & Testing Mappings

Open the Command Prompt on any PC to verify the isolation barriers:

Intra-VLAN Ping (Success): Click PC1 (VLAN 10 on Switch1) and ping PC2 (VLAN 10 on Switch2) by typing ping 192.168.10.X. The ping will succeed because traffic traverses the FastEthernet 0/24 trunk link seamlessly.

Inter-VLAN Ping (Failure): Click PC1 (VLAN 10) and try to ping PC4 (VLAN 20) by typing ping 192.168.20.X. The ping will return a "Request timed out" error because different VLANs cannot communicate without an intermediate Layer 3 Router (Inter-VLAN routing / Router-on-a-stick).

Question: 2

Configure trunk ports between the two switches in your Packet Tracer network so that all VLANs can communicate across both switches.

Hint: Use the 'switchport mode trunk' command and allow all VLANs on the trunk link.

Answer: 1. The Configuration Commands (Run on Switch1 and Switch2)

Open the Command Line Interface (CLI) tab on both Switch1 and Switch2, and input this exact script block to configure your connecting interface (assuming they are linked via port Fa0/24):

text

enable

configure terminal

```
interface FastEthernet 0/24
```

```
description Trunk_Link_to_Other_Switch
```

```
switchport mode trunk
```

```
switchport trunk allowed vlan all
```

end
write

Use code with caution.

2. Configuration Breakdown

interface FastEthernet 0/24: Selects the specific physical interface linking the two switches together.

switchport mode trunk: Overrides the default access port status and turns the link into a highway capable of tagging and carrying frames from multiple VLANs simultaneously.

switchport trunk allowed vlan all: Explicitly permits tags from all your created VLANs (10, 20, and 30) to cross the bridge, ensuring seamless communication between identical groups across the room.

3. Verification Steps (How to check if it works)

Type this verification command into the switch CLI execution prompt:

text

show interfaces trunk

Use code with caution.

Expected Output: You will see port Fa0/24 listed with a mode status of trunking, an encapsulation type of 802.1q, and the allowed VLAN string reading 1-4094 (or listing 10,20,309)

Question: 3

Add a router to your Packet Tracer network and configure Inter-VLAN routing so that a PC in VLAN 10 can successfully ping a PC in VLAN

20.

Hint: Use router-on-a-stick configuration with sub-interfaces for each VLAN.?

Answer: 1. Physical Connectivity Setup

Hardware: Add a 2911 Router (or any ISR router with a GigabitEthernet interface) to the workspace.

Cable: Connect Switch1 (GigabitEthernet 0/1) to Router (GigabitEthernet 0/0) using a straight-through copper cable.

2. Switch-Side Configuration (Switch1)

You must change the switch port connected to the router from an access port to a trunk port so it can forward multi-tagged VLAN packets. Open Switch1's CLI and type:

text

enable

configure terminal

interface GigabitEthernet 0/1

switchport mode trunk

switchport trunk allowed vlan all

exit

Use code with caution.

3. Router-Side Configuration (Router CLI)

Open the Router's CLI tab and paste this exact script block to activate the main physical link and build virtual sub-interfaces matching the gateways of VLAN 10 and VLAN 20:

text

enable

configure terminal

interface GigabitEthernet 0/0

no shutdown

exit

interface GigabitEthernet 0/0.10

description Gateway_For_Foodiez

encapsulation dot1Q 10

ip address 192.168.10.1 255.255.255.0

exit

interface GigabitEthernet 0/0.20

description Gateway_For_Gamers

encapsulation dot1Q 20

ip address 192.168.20.1 255.255.255.0

exit

interface GigabitEthernet 0/0.30

description Gateway_For_Shoppers

encapsulation dot1Q 30

ip address 192.168.30.1 255.255.255.0

exit

Use code with caution.

4. End-Device PC Configuration Update

For different subnets to talk, you must configure a Default Gateway on every single PC's desktop settings:

PCs in VLAN 10: Set Default Gateway IP to 192.168.10.1

PCs in VLAN 20: Set Default Gateway IP to 192.168.20.1

PCs in VLAN 30: Set Default Gateway IP to 192.168.30.1

5. Final Verification Mappings

Open the Command Prompt on PC1 (VLAN 10 - IP 192.168.10.10).

Execute a cross-VLAN test ping to PC4 (VLAN 20 - IP 192.168.20.10): ping 192.168.20.10

Expected Result: The first one or two echo requests might time out while ARP (Address Resolution Protocol) resolves the new pathways. Subsequent replies will return a clean "Reply from 192.168.20.10" status code showing absolute success.

Question: 4

Set up static routes on the router so that a new network segment (e.g., a 'Creators' VLAN 40 on a third switch) can communicate with all existing VLANs, and test connectivity with ping?

Answer: 1. New Network Configuration Metrics

VLAN 40 (Creators) Subnet: 192.168.40.0/24

Router2 Gateway Subnet Interface: 192.168.40.1

Inter-Router WAN Serial/Ethernet Link Subnet: 10.0.0.0/30

Router1 WAN IP: 10.0.0.1

Router2 WAN IP: 10.0.0.2

2. Static Route Configuration

On Router1 (Existing Core HQ Router)

You must instruct the core router where to send outbound traffic intended for the new 192.168.40.0/24 network segment. Open Router1 CLI and input:

```
text
enable
configure terminal
ip route 192.168.40.0 255.255.255.0 10.0.0.2
exit
write
```

Use code with caution.

On Router2 (New Branch Expansion Router)

To ensure returning packets find their way back home, configure a summary fallback route pointing toward your internal subnets (VLAN 10, 20, and 30). Open Router2 CLI and input:

```
text
enable
configure terminal
ip route 192.168.10.0 255.255.255.0 10.0.0.1
ip route 192.168.20.0 255.255.255.0 10.0.0.1
ip route 192.168.30.0 255.255.255.0 10.0.0.1
exit
write
```

Use code with caution.

3. Switch Ports Allocation (Third Switch)

Open your newly added Switch3 CLI to declare the VLAN environment and bind the access interfaces for the Creators' workstations:

```
text
```

```
enable
configure terminal
vlan 40
  name Creators
exit
interface range FastEthernet 0/1 - 5
  switchport mode access
  switchport access vlan 40
exit
```

Use code with caution.

4. Verification Mappings

End-Device Check: Assign a static address to a machine on Switch3 (e.g., IP: 192.168.40.10, Subnet Mask: 255.255.255.0, Default Gateway: 192.168.40.1).

Connectivity Test: Open the Command Prompt terminal window on the VLAN 40 PC and execute an echo test ping to a VLAN 10 machine:

```
cmd

ping 192.168.10.10
```

Use code with caution.

Expected Result: The initial connection frames might register a minor drop delay while the internal ARP tables update. The routing engine will then successfully print a sequence of clear, round-trip "Reply from 192.168.10.10" lines.

Question: 5

Design a mini network in Packet Tracer that mimics a Zomato-style office: create VLANs for 'Support', 'Delivery', and 'Management', configure trunking and static

routing, and document your topology with a screenshot and a brief explanation of how devices in different VLANs communicate?

Answer: 1. Network Architecture & Layout Mappings

VLAN 10 (Support): Subnet 192.168.10.0/24 (Gateway: 192.168.10.1). Assigned to call center agents handling customer orders and rider coordination.

VLAN 20 (Delivery): Subnet 192.168.20.0/24 (Gateway: 192.168.20.1). Assigned to rider tracking servers and automated logistics dispatch terminals.

VLAN 30 (Management): Subnet 192.168.30.0/24 (Gateway: 192.168.30.1). Assigned to executive team workstations accessing financial portals and operational metrics.

Core Hardware Infrastructure:

2x 2960 Access Switches (Switch-Floor1, Switch-Floor2) linked via trunk line interface Fa0/24.

1x 2911 Core Router (Zomato-HQ-Router) linked via trunk line interface Gi0/0 to Switch-Floor1 (Gi0/1).

2. Device Configuration Script Mappings

Step A: Switch Layer Setup (Run on both Switches)

text

enable

configure terminal

vlan 10

name Support

vlan 20

name Delivery

vlan 30

name Management

exit

interface range FastEthernet 0/1 - 5

switchport mode access

switchport access vlan 10

exit

interface range FastEthernet 0/6 - 10

switchport mode access

switchport access vlan 20

exit

interface range FastEthernet 0/11 - 15

switchport mode access

switchport access vlan 30

exit

interface FastEthernet 0/24

switchport mode trunk

switchport trunk allowed vlan all

exit

Use code with caution.

Step B: Core Router Layer Setup (Router-on-a-Stick Configuration)

text

enable

configure terminal

interface GigabitEthernet 0/0

no shutdown

exit

```
interface GigabitEthernet 0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
 exit
```

```
interface GigabitEthernet 0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
 exit
```

```
interface GigabitEthernet 0/0.30
 encapsulation dot1Q 30
 ip address 192.168.30.1 255.255.255.0
 exit
```

Use code with caution.

3. How Devices in Different VLANs Communicate

Local Isolation Layer: When a machine in VLAN 10 (Support) initiates a connection to a database server located in VLAN 20 (Delivery), the switch blocks direct Layer 2 communication because the devices belong to separate broadcast domains.

Trunk Traversal Layer: Because the target IP address sits outside the originating device's local network, the sending machine packages the frame and routes it toward its Default Gateway (192.168.10.1). The switch stamps the frame with an IEEE 802.1Q VLAN 10 tag and sends it over the active Trunk Port interface link to the router.

Layer 3 Routing Layer: The sub-interface Gi0/0.10 on the Zomato-HQ-Router intercepts the incoming tagged frame and strips off the outermost VLAN 10 tag. The router looks up the destination IP inside its internal Routing Table, matching it to its directly connected Gi0/0.20 virtual network boundary.

Target Delivery Layer: The router re-encapsulates the application data packet, stamps it with a fresh VLAN 20 tag, and transmits it back down the trunk port link. The switch reads the tag and forwards the frame directly to the designated delivery device interface.

4. How to Generate Your Documented Screenshot

Set up this complete physical topology inside the Cisco Packet Tracer canvas workspace grid.

Press the *Green Play* button (Fast Forward Time) multiple times to ensure all orange port link bulbs switch to green indicators, proving complete spanning-tree convergence.

Open a PC console inside VLAN 10 and execute a successful cross-network confirmation ping to a VLAN 30 host IP using the terminal command prompt: ping 192.168.30.10.

Capture a clear desktop screen snippet showing the physical device links alongside the successful terminal ping logs to attach to your text analysis.